



Document Management System



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Title: Nutrition Policy in PICU, PCSU and PHDU

This Policy is not to be applied on patients with nephrologic and metabolic disorders; since these patients have their own nutritional requirement, which is not included in this policy.

The pediatric intensive care unit (PICU) environment faces unique challenges to achieve enteral nutrition (EN) goals for the critically ill child. Since decades complexity of care provided to acutely ill children in PICU have often come across conflicts with nutritional support.

Children (ages 1 month to 13 years) admitted to the pediatric intensive care unit (PICU) may already have some degree of malnutrition. In addition, critical illness places variable energy demands on the child that cannot always be accurately predicted using standard equations for calculating energy expenditure. (2'3)

To achieve adequate nutritional support every PICU requires a policy and pathway to follow. By this targeted goal can be achieved with great outcomes.

Our Targets of nutritional support in critical care areas:

1. Prevent Protein Catabolism.
2. Prevent under and over feeding.
3. Achieve normal growth.

Elements of critical care

Enteral nutrition (EN) practice bundle⁽¹⁾

1. Nutritional assessment on admission
2. Caloric goal (measured by energy expenditure or estimate)
3. Early enteral nutrition (within 24 hours of admission)

5. Daily checklist for enteral nutrition status/protocolized EN advancement

PICU Enteral Nutrition Support Algorithm

Oral route unavailable OR Unable to protect airway

Nutrition assessment, Weight on admission Identify caloric goal and referral to dietician

HEAD OF BED elevated 30° unless contraindicated

EVALUATE FOR RISK OF ASPIRATION

(Depressed gag/cough, altered sensorium, delayed gastric emptying, GE reflux, severe bronchospasm, history of reflux)

Aspiration risk

No aspiration risk

TRANSPYLORIC (Nasojejunal / Gastrojejunal) FEEDINGS with gastric decompression

NASOGASTRIC / GASTRIC TUBE FEEDINGS
BS present / No gastric distension

Abnormal
Gastric emptying

Normal
Gastric emptying

Continuous Feeding

Bolus Feeding

GOAL CALORIES START:
1-2 mL/kg/hr
(max 25 mL/hr)
(OR 0.5 mL/kg/hr if risk of gut ischemia)
ADVANCE:
<1 yr: 1-5 mL/hr q4hrs
>1 yr: 5-20 mL/hr q4hrs
until goal reached

TROPIC FEEDING
0.5 mL/kg/hr
OR
20 mL/hr
Full-strength formula or breast milk

GOAL CALORIES START:
25% of Goal volume or full feed if patient if less than 24 hours of admission, (Full-strength feeding volume required to reach caloric goal). Bolus feedings divided q3hr (if <6 months of age) or q4hr (if >6 months of age).
If volume intolerant, try smaller volumes more frequently or continuous feedings.
ADVANCE:
Increase each feeding by 25% volume until goal reached

ENTERAL FEEDING INTOLERANCE:

Gastric residual volumes (GRV) recorded prior to each bolus feed or q4hrs in patients on continuous gastric feedings with abdominal discomfort, distention or emesis.
 If GRV >150 mL; or 5 mL/kg, or >½ volume of previous feeding; or >total 2 hourly infusion rate in patients on continuous feeding – hold feedings and repeat GRV after 2 hrs. If repeat GRV is elevated, hold feedings and monitor GRV at 4 hrs.
 If abdominal distention, (abdominal girth increased for 2 consecutive measurements) or abdominal discomfort or emesis x2 – hold feedings for 4 hrs and reassess.

Pressors and Nutrition :

Enteral feedings should not be initiated if patient is on:

- Two inotropes
- One inotrope (adrenaline > 0.05 or noradrenaline > 0.05 or Dopamine > 10)

Respiratory quotient (RQ) may help determine substrate utilization (fat, protein, mixed, carbohydrate or fat synthesis). Patients that are spontaneously breathing are often measured under a hood which creates a seal to trap all oxygen inhaled and carbon dioxide exhaled. Ventilated patients or patients with tracheostomies may also be tested, as long as there are no inspiratory or expiratory air leaks and the patient is not on high frequency ventilation of FIO₂ of >60%.

RQ <0.85 indicates underfeeding

RQ 0.85-1.0 indicates adequate feeding

RQ >1.0 indicates overfeeding

The respiratory quotient (RQ) measures the ratio of the volume of carbon dioxide (V_c) produced by an organism to the volume of oxygen consumed (V_o). This is represented by the following equation:

$$RQ = \frac{V_c}{V_o}$$

Determining Calorie and Protein Needs in PICU

- Do **not** use the standard **RDA tables** initially to calculate energy requirements.
- Estimate resting energy expenditure (REE)
- Determine Stress Factor

(WHO and Schofield)

Total Calories = REE X Activity Factor (AF) X Stress Factor (SF)

- Estimate patient's protein requirements

Total Protein = Protein RDAs X Stress Factor

Stress Factors:

Clinical Condition

Fever	12% per degree > 37 C
Routine/elective surgery, minor sepsis	1.1 – 1.3
Cardiac failure	1.25 – 1.5
Major surgery	1.2 – 1.5
Sepsis	1.4 – 1.6
Catch-up Growth	1.5 – 2.0
Trauma or head injury	1.5 – 1.8
Starvation	0.7
Burn	1.5 – 2.5

Activity Factors:

Paralyzed	1.0
Confined to bed	1.1
Ambulatory	1.2 – 1.3

Resting Energy Expenditure

Age	REE (Kcal/Kg)
0-6 months	55
7-12 months	55
1-3 years	57
4-6 years	48

11-13 years (M/F)	32/28

Example: 6 month old intubated sedated only, with septic shock temperature 36

Total Calories (KCL/Kg/day) = REE X AF X SF

55 X 1.1 X 1.6 = 97 Kcl/Kg/day (100 Kcl/Kg/day)

Protein Requirement (RDA)

Age	Protein (gm/kg)
0-6 months	2.2
7-12 months	1.6
1-3 years	1.2
4-6 years	1.2
7-10 years	1.0
11-13 years (M/F)	1.0/0.8

Complications of Overfeeding:

- Secretory diarrhea
- Hyperglycemia (may increase infection)
- Lipogenesis with increased insulin production
- Hepatic complications: fatty liver, intrahepatic cholestasis
- Electrolyte abnormalities: ↓ PO₄, ↓ K, ↓ Mg
- Volume overload,
- ↑ CO₂ production- ↑ ventilatory demand
- ↑ O₂ consumption
- Pulmonary edema & respiratory failure

NUTRITION CHECKLIST

Sticker

Weight on admission

Age

DOA

Ordered by: ☐ Doctor ☐ Dietician

Day	IVF	Feeding	weight	Types of formula	Remarks	Name/ Sign
Day 1 Date:						
Day 2 Date:						
Day 3 Date:						
Day 4 Date:						
Day 5 Date:						
Day 6 Date:						
Day 7 Date:						
Day 8 Date:						
Day 9 Date:						
Day10 Date:						
Day 11 Date:						
Day 12 Date:						

Prepared by:

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Reference:

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